- Hasta ahora hemos estado estudiando ecuaciones de este tipo $\begin{cases} u'' = f(t,u,u') \\ u(t_0) = u_0 \\ u'(t_0) = v_0 \end{cases}$

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$u'' = g(t,u,u')$ $\quad g: \mathbb{R}\times\mathbb{R}^2 \longrightarrow \mathbb{R}$ continua, $\frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}$ continuas

$(t,x,y) \longmapsto g(t,x,y)$

$C^2(J,\mathbb{R})$

$u''(t) = g(t,u(t),u'(t))$

sistema de primer orden

$\begin{cases} u' = v \\ v' = g(t,u,v) \end{cases}$ $\quad \Sigma_e \longmapsto \Sigma_{s,1}$

$u \longmapsto D(u) = \begin{pmatrix} u \\ u' \end{pmatrix} = \begin{pmatrix} u \\ v \end{pmatrix}$ biyección

$\begin{pmatrix} u \\ v \end{pmatrix}' = \begin{pmatrix} v \\ g(t,u,v) \end{pmatrix} = f(t,u,v)$ Localmente Lipschitz.

$Df(t,u,v) = \begin{pmatrix} 0 & 1 \\ \frac{\partial g}{\partial u} & \frac{\partial g}{\partial v} \end{pmatrix}$ $\quad u(t_0) = u_0$, $v(t_0) = v_0$

- Si fuera de tercer orden:

$u''' = g(t,u,u',u'')$ $\quad g: \mathbb{R}\times\mathbb{R}^3 \longrightarrow \mathbb{R}$ continua, $\frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}, \frac{\partial g}{\partial z}$ continuas

$(t,x,y,z) \longmapsto g(t,x,y,z)$

$\begin{cases} u' = v \\ v' = w \\ w' = g(t,u,v,w) \end{cases}$ $\quad u \longmapsto D(u) = \begin{pmatrix} u \\ u' \\ u'' \end{pmatrix}$ biyección

$\begin{pmatrix} u \\ v \\ w \end{pmatrix}' = \begin{pmatrix} v \\ w \\ g(t,u,v,w) \end{pmatrix} = f(t,u,v,w)$ Localmente Lipschitz

$Df(t,u,v,w) = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ \frac{\partial g}{\partial u} & \frac{\partial g}{\partial v} & \frac{\partial g}{\partial w} \end{pmatrix}$ $\quad (u(t_0), v(t_0), w(t_0)) = (u_0, v_0, w_0)$

- Para orden N:

$u^{(N)} = g(t,u,u',\dots,u^{(N-1)})$ $\quad g: \mathbb{R}\times\mathbb{R}^N \longrightarrow \mathbb{R}$

$(t,x) \longmapsto g(t,x_1,x_2,\dots,x_N)$ continua

$\frac{\partial g}{\partial x_j}$ continuas $\quad 1 \le j \le N$

$u_1 = u$
$u_2 = u'$
$\vdots$
$u_N = u^{(N-1)}$

$\begin{pmatrix} u_1 \\ \vdots \\ u_N \end{pmatrix} = D(u) = \begin{pmatrix} u \\ u' \\ \vdots \\ u^{(N-1)} \end{pmatrix}$, $\begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_N \end{pmatrix}' = \begin{pmatrix} u_2 \\ u_3 \\ \vdots \\ u_N \\ g(t,u_1,\dots,u_N) \end{pmatrix} = f(t,u_1,\dots,u_N)$

$Df(t,u_1,\dots,u_N) = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ \frac{\partial g}{\partial u_1} & \frac{\partial g}{\partial u_2} & \frac{\partial g}{\partial u_3} & \cdots & \frac{\partial g}{\partial u_N} \end{pmatrix}$ $\quad (u_1(t_0), u_2(t_0), \dots, u_N(t_0)) = (u_{1,0}, \dots, u_{N,0}) \in \mathbb{R}^N$

Podemos aplicar toda la teoría a sistemas de este estilo

- Si tuviéramos un sistema de este estilo (que podrían ser dos péndulos, uno sujeto a otro)

$$\begin{cases} u'' = g(t,u,u',v,v') \\ v'' = h(t,u,u',v,v') \end{cases} \longrightarrow \text{Sistema de primer orden asociado} \longrightarrow \begin{cases} u' = z \\ z' = g(t,u,z,v,w) \\ v' = w \\ w' = h(t,u,z,v,w) \end{cases}$$

$u(t_0) = u_{0,0}$, $u'(t_0) = u_{0,1}$

$v(t_0) = v_{0,0}$, $v'(t_0) = v_{0,1}$

$$f(t,u,z,v,w) = \begin{pmatrix} z \\ g(t,u,z,v,w) \\ w \\ h(t,u,z,v,w) \end{pmatrix} \qquad Df(t,u,z,v,w) = \begin{pmatrix} 0 & 1 & 0 & 0 \\ \frac{\partial g}{\partial u} & \frac{\partial g}{\partial z} & \frac{\partial g}{\partial v} & \frac{\partial g}{\partial w} \\ 0 & 0 & 0 & 1 \\ \frac{\partial h}{\partial u} & \frac{\partial h}{\partial z} & \frac{\partial h}{\partial v} & \frac{\partial h}{\partial w} \end{pmatrix}$$

podemos aplicar toda la teoría

$$\begin{cases} u' = z \\ z' = g(t,u,z,v,w) \\ v' = w \\ w' = h(t,u,z,v,w) \end{cases} \longrightarrow (I, u, z, v, w) \longrightarrow (I, u, v)$$

son redundantes

$u(t_0) = u_0$, $z(t_0) = z_0$, $v(t_0) = v_0$, $w(t_0) = w_0$